Role of u_s and Exact-vs-Bounded Analysis Modes in First-Step-Contraction MPC

April 6, 2026

1 Purpose

This note answers two related questions for the current first-step-contraction MPC path:

1. Is the steady-state input target u_s actually used in the Lyapunov logic, or is only x_s relevant?
2. Why can the offline exact frozen-$\hat{d}$ target look more "correct" than the bounded target in some plots, even though the online controller uses the bounded target?

The short answer is:

- the exact frozen-$\hat{d}$ target is the correct *unconstrained model equilibrium diagnostic*,
- the bounded frozen-$\hat{d}$ target is the correct *feasible controller target* when the exact target violates the input box,
- in the maintained first-step-replacement path the hard first-step contraction constraint is centered on x_s, not directly on u_s,
- but u_s is still used in Lyapunov-related prediction and in the standard tracking formulation.

2 Scaled Deviation Coordinates

All calculations below are in the same scaled deviation coordinates used by the identified linear model, the observer, and the offset-free MPC code. Let

$$u_k = u_k^{\text{scaled}} - u_{ss}^{\text{scaled}}, \tag{1}$$

$$y_k = y_k^{\text{scaled}} - y_{ss}^{\text{scaled}}. \tag{2}$$

The observer estimate, the target package, the Lyapunov center, and the controller action all live in these deviation coordinates.

3 Augmented Observer and Frozen-Disturbance Target

The offset-free observer produces the augmented estimate

$$\hat{z}_k = \begin{bmatrix} \hat{x}_k \\ \hat{d}_k \end{bmatrix}, \tag{3}$$

where $\hat{x}_k \in \mathbb{R}^{n_x}$ is the physical-state estimate and $\hat{d}_k \in \mathbb{R}^{n_y}$ is the output-disturbance estimate.

The bounded frozen-$\hat{d}$ target method freezes the disturbance target at

$$d_s(k) = \hat{d}_k. \tag{4}$$

Then the steady-state equations use the unaugmented plant matrices (A, B, C):

$$x_s = Ax_s + Bu_s, \tag{5}$$
$$y_s = Cx_s + d_s. \tag{6}$$

If the requested output target is $y_{sp,k}$, then the exact offset-free steady-state equations are

$$(I - A)x_s - Bu_s = 0, \tag{7}$$
$$Cx_s = r_k, \tag{8}$$

with

$$r_k := y_{sp,k} - \hat{d}_k. \tag{9}$$

4 Exact Unbounded Target

Define

$$M := \begin{bmatrix} I - A & -B \\ C & 0 \end{bmatrix}, \qquad w_s := \begin{bmatrix} x_s \\ u_s \end{bmatrix}, \qquad b_k := \begin{bmatrix} 0 \\ r_k \end{bmatrix}. \tag{10}$$

Then the exact target solves

$$Mw_s = b_k. \tag{11}$$

If $(I - A)$ is invertible, this can be reduced to

$$Gu_s = r_k, \qquad G := C(I - A)^{-1}B, \tag{12}$$

with

$$x_s = (I - A)^{-1}Bu_s. \tag{13}$$

This exact target is the correct frozen-disturbance equilibrium of the linear model. If the purpose is to answer

> "What stationary input-state pair would the model need in order to maintain the set-point under the current disturbance estimate?"

then the exact target is the correct answer.

5 Bounded Feasible Target

The exact target may be infeasible because u_s may violate the controller box

$$u_{\min} \le u_s \le u_{\max}. \tag{14}$$

When that happens, the exact target is still valuable as a diagnostic, but it is not a feasible controller center.

The bounded frozen-$\hat{d}$ fallback solves

$$\min_{u_s} \quad \|Gu_s - r_k\|_2^2 + \lambda_u \|u_s - u_{\mathrm{ref},k}\|_2^2 \qquad \text{subject to } u_{\min} \le u_s \le u_{\max}, \tag{15}$$

where in the current notebook

$$u_{\mathrm{ref},k} = u_{k-1}, \tag{16}$$

and $\lambda_u \ge 0$ is the user-set weight exposed in the notebook.

After the bounded input target is found, the corresponding state target is

$$x_s = (I - A)^{-1} B u_s, \tag{17}$$

or the equivalent stacked least-squares state if the reduced form is not used.

This bounded target is the correct answer to a different question:

> "What feasible steady-state pair is closest to the frozen-disturbance equilibrium while respecting the input box and optional anchoring to the previous input?"

6 Why Exact and Bounded State Targets Can Look Very Different

Suppose the exact input target is outside the box. Then the bounded optimization pushes u_s back to the feasible set. Because x_s and u_s are coupled through

$$(I - A)x_s = Bu_s, \tag{18}$$

the state target changes as well. Therefore it is completely possible for

$$x_s^{\mathrm{exact}} \ne x_s^{\mathrm{bounded}} \tag{19}$$

by a large amount.

This is why an offline exact plot can look "more correct" as a pure equilibrium diagnostic: it is the exact solution of the frozen-$\hat{d}$ equations. But if that target requires infeasible inputs, it is not the same object as the feasible target used online.

7 Where u_s Enters the Lyapunov Logic

7.1 Physical Error Centering

The physical-state tracking error is defined around the target state:

$$e_x = \hat{x}_k - x_s. \tag{20}$$

So every Lyapunov quantity is centered at x_s.

7.2 One-Step Predicted Error

The code also uses the input target when predicting the next physical error:

$$e_x^+ = Ae_x + B(u - u_s). \tag{21}$$

This identity comes from subtracting the target steady-state dynamics

$$x_s = Ax_s + Bu_s \tag{22}$$

from the plant prediction

$$x^+ = A\hat{x}_k + Bu. \tag{23}$$

Therefore u_s is not an incidental quantity. It participates in the Lyapunov-centered error propagation.

7.3 Candidate Evaluation

When the upstream MPC proposes a candidate input u_{cand}, the candidate evaluation computes

$$e_x = \hat{x}_k - x_s, \tag{24}$$
$$e_x^+ = Ae_x + B(u_{\text{cand}} - u_s), \tag{25}$$
$$V_k = e_x^\top P_x e_x, \tag{26}$$
$$V_{k+1}^{\text{cand}} = (e_x^+)^\top P_x e_x^+. \tag{27}$$

Then it checks the Lyapunov inequality

$$V_{k+1}^{\text{cand}} \leq \rho V_k + \varepsilon. \tag{28}$$

So in the candidate-evaluation step, u_s explicitly affects the predicted next Lyapunov value.

7.4 Hard First-Step-Constrained Replacement

The maintained first-step replacement path solves a constrained upstream MPC candidate problem whose additional hard constraint is

$$V(x_{1|k} - x_s) \leq \rho V(\hat{x}_k - x_s) + \varepsilon. \tag{29}$$

This constraint is written directly in terms of predicted state and target state:

$$(x_{1|k} - x_s)^\top P_x (x_{1|k} - x_s) \leq \rho(\hat{x}_k - x_s)^\top P_x (\hat{x}_k - x_s) + \varepsilon. \tag{30}$$

In this hard first-step inequality, u_s does not appear explicitly. The online constrained replacement is centered on x_s.

This is the most important structural answer:

> for the current first-step-replacement path, the hard Lyapunov contraction constraint depends directly on x_s, while u_s enters indirectly through the steady-state relation and through separate candidate-evaluation logic.

7.5 Standard Tracking Lyapunov MPC Formulation

The repository also contains the full standard tracking Lyapunov formulation, where the cost includes input tracking terms around u_s:

$$\sum_{i=0}^{N_c-1} (u_{k+i} - u_s)^\top S_u (u_{k+i} - u_s), \tag{31}$$

along with state centering around x_s and the terminal Lyapunov term

$$(x_{k+N} - x_s)^\top P_x (x_{k+N} - x_s). \tag{32}$$

So for the general Lyapunov-tracking formulation, u_s is absolutely part of the control design. It is only the specific hard first-step replacement inequality that is written entirely around x_s.

8 Implication for Analysis Plots

The offline sidecar now supports three analysis modes:

- **exact**: use the unbounded frozen-$\hat{d}$ target everywhere in the diagnostic plots,
- **bounded**: use the bounded feasible frozen-$\hat{d}$ target everywhere in the diagnostic plots,
- **hybrid**: preserve the older mixed view, where state/output/disturbance use the exact target while input-side plots use the bounded target.

These modes are *analysis-only*. They do not change the online controller unless the online target-generation mode itself is changed.

9 Why the Exact Offline Plot Can Be the Better Diagnostic

If your goal is to inspect model consistency, then the exact mode is often the better diagnostic because it answers:

$$Mw_s = b_k \tag{33}$$

exactly. In that view,

- if $r_k \to 0$, then the exact target tends to the nominal equilibrium,
- if $\hat{x}_k$ is converging to the frozen-disturbance steady-state manifold, then $\hat{x}_k$ should move toward x_s^{exact}.

If instead your goal is to inspect what the controller can feasibly aim for under box constraints, then bounded mode is the better diagnostic.

10 Practical Interpretation

There are therefore two legitimate but different objects:

$$(x_s^{\text{exact}}, u_s^{\text{exact}}) : \quad \text{model-equilibrium diagnostic}, \tag{34}$$

$$(x_s^{\text{bounded}}, u_s^{\text{bounded}}) : \quad \text{feasible controller target}. \tag{35}$$

Comparing the online controller trajectory to the exact target is useful for understanding the unconstrained frozen-disturbance equilibrium. Comparing it to the bounded target is useful for understanding what the constrained controller can actually use as a center.

11 Answer to the Original Question

If the question is:

> "Are we even using u_s in the Lyapunov logic at all?"

then the precise answer is:

- **Yes**, u_s is used in Lyapunov-centered one-step prediction and candidate evaluation through $e_x^+ = Ae_x + B(u - u_s)$.
- **Yes**, u_s is used in the full standard Lyapunov tracking MPC objective through input-centering penalties.
- **No**, the current hard first-step contraction constraint itself is written only around x_s and does not contain u_s explicitly.

So if the offline exact state target looks more physically meaningful for diagnosis, that does not contradict the current online implementation. It just means the exact target is the better unconstrained equilibrium diagnostic, while the bounded target remains the feasible control target.